Greecon
SH.P.K

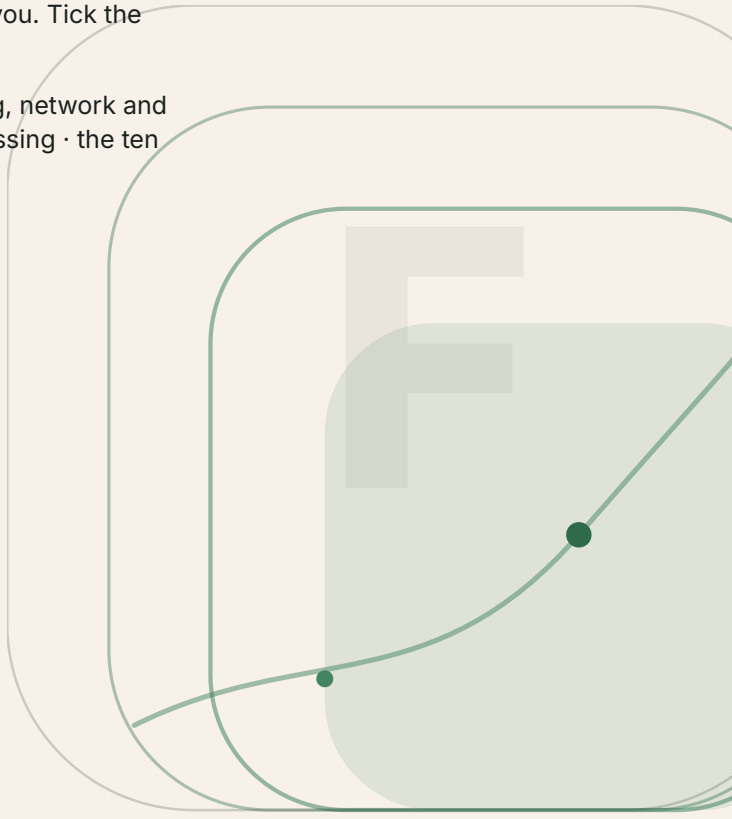
F

SITE PACK F

Farm or greenhouse with irrigation

Soil and weather, with the water supply behind it. Everything for one kind of site in one pack: what to buy, the survey, the diagrams, install day step by step, the platform steps, the commissioning tests and the handover sheet.

FOR	The project manager at this kind of site, the electrician, and the customer's owner at handover
PRINT	The whole pack, for the site in front of you. Tick the boxes as you go
COVERS	Bill of materials · survey · system, wiring, network and enclosure diagrams · install day · addressing · the ten setup steps · commissioning · handover



Version 1.0 · 27 October 2026

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SITE PACK F

Farm or greenhouse with irrigation

Soil and weather, with the water supply behind it. A complete pack for one kind of site, in the order of the work: what to buy, the survey, the drawings, the office, install day step by step, the platform's setup list, the commissioning tests and the handover. Print all of it and tick the boxes as you go. The electrician does everything on the mains side; you do the low voltage and the software; Greecon only reads.

At a glance

WHO IT SUITS
A farm or a greenhouse with irrigation: soil moisture and weather at the field, with the water supply behind it (a tank, a pump, the pressure in the pipe).

WHAT IT MEASURES
The moisture of the soil (one probe for each zone), the air temperature and humidity, the tank, the line pressure and, if there is a pump to watch, the pump's flow and whether it runs.

PLANNING ESTIMATE
About EUR 700 to 1,500.

TIME
Survey about 1 hour, with the grower. Install day 4 to 6 hours. Commissioning about 1 hour. Handover 30 minutes.

TEMPLATE IN THE PLATFORM
Farm or greenhouse with irrigation

STANDARD SETTINGS
Water module unit 3. 9600 baud, 8 data bits, no parity, 1 stop bit. Converter on port 502, in Modbus TCP to RTU gateway mode. A cellular router is the likelier case in a field.

STOP

Greecon gives no irrigation advice until the grower's own "start irrigating below" value is entered. That number depends on the soil, the crop and the probe's depth, and nobody can set it for the grower. After the first week of readings, ask the grower what reading means "time to water", and enter it as the limit (step 9 of the setup list). **Frost works the same way:** ask at what air temperature frost damages the crop, and in which weeks; enter it as **Frost alert at or below** on the air temperature (step 9). It opens an Act now alert, so agree who is woken. Map the air temperature as a signed number (int16), or a frosty morning reads as a hot afternoon.

The order of the work: survey with the grower, buy, prepare at the office, install day, the platform's setup list, commissioning, handover. Then the morning check, and the day 1, 3 and 7 calls.

STOP

Six rules for the whole job.

1. **Greecon only reads.** It measures; it never switches anything. Never connect a wire from Greecon's equipment to a pump starter, an inverter's control, a valve or any safety circuit.
2. **Anything inside a distribution board is the electrician's work,** under the customer's lock-out procedure, with the project manager watching and writing it down. Nobody works on a live conductor.
3. **The project manager does the low-voltage and software side only:** the box, the network, the converter, the data cable (run to the board; the electrician connects its ends at the meters, because their terminals are inside the board), the 4 to 20 mA signal wires, and the 12 or 24 V supply from a plug-in supply.
4. **Never touch the solar panels' wiring (DC), a tank you must climb into, wastewater or chemicals.** The customer's own staff or installer does that. If the only place for a sensor is there, say it is not possible and write why.
5. **Greecon is not a safety system.** The protection already in the customer's equipment (float switches, relief valves, trips) stays the safety layer.
6. **Write everything down at the time.** Photograph every meter's label and display, the wiring before the board is closed, the converter's settings and the box's label.

How to read this pack

Mark	What it means
ELECTRICIAN	The electrician. Everything inside a distribution board, every current clamp and voltage tap, and the pump's run contact.
PROJECT MANAGER	The project manager: low voltage and software.
CUSTOMER	The customer: access, people, and starting or stopping their own equipment.
☐	A box to tick when it is done. Print the pack and tick as you go.
Do, **Check**, **Stop**, **Tip**, **Say**	The shaded boxes: do this; check this; stop and call the electrician; a useful habit; words to say to the customer.
Figure F-1 and so on	Drawings of this kit. Every drawing is marked with the kit it is for. Terminal numbers are never drawn: follow the maker's manual.

What it measures

The template **Farm or greenhouse with irrigation** makes three devices. What each gives:

Reading in Greecon	Comes from	Unit	Standard alert limits
Soil moisture	Soil probe: a soil moisture probe that speaks RS-485 Modbus, one for each zone, unit number chosen and written down	%	None. **Start irrigating below** has no standard value: the grower's own is entered after the first week
Air temperature and **Humidity**	Weather sensor: a temperature and humidity sensor that speaks RS-485 Modbus, unit number chosen and written down	degrees Celsius, %	None
Tank level	Water I/O module, unit 3: the level transmitter's 4 to 20 mA	%	Critical below 15 % (Act now); Low below 35 % (Check today)
Line pressure	Water I/O module, unit 3: the pressure transmitter's 4 to 20 mA	bar	None
Pump flow (if there is a pump to watch)	Water I/O module, unit 3: the flow meter's 4 to 20 mA	litres a minute	No flow at or below 0.2 litres a minute (Act now, once it has held for 30 seconds, and only when the box reads both a flow and the pump's running state)
Pump running (if there is a pump to watch)	Water I/O module, unit 3: the pump starter's spare contact	on or off	None

A sensor whose maker does not document its registers cannot be mapped, so it cannot be used. Soil readings are only as good as the contact between the probe and the soil: the handover records the probe's depth and where it is.

What to buy and bring

	Item	Specification	Qty	Planning estimate (EUR)
☐	**Box**	Fanless x86 mini PC (see the box specification), 8 GB, 128 GB or more SSD or eMMC, 12 to 24 V DC input	1	180 to 300
☐	**Converter**	RS-485 to Ethernet, DIN rail, 9 to 36 V DC, Modbus TCP to RTU gateway mode	1	30 to 60 (basic) or 250 and up (industrial)
☐	**Analog input module**	Modbus RTU, at least 4 channels, 4 to 20 mA, 7 to 36 V DC	1	30 to 60 (basic) or 150 to 250 (industrial)

	Item	Specification	Qty	Planning estimate (EUR)
☐	**Level transmitter**	Submersible, vented cable, 4 to 20 mA, range at least the tank's height, wetted parts approved for drinking water (or a non-contact ultrasonic sensor above the tank)	1	120 to 400
☐	**Pressure transmitter**	4 to 20 mA, range to suit the pipe (0 to 10 bar is usual), stainless wetted parts	1	40 to 150
☐	**24 V DIN power supply**	30 to 60 W	1	25 to 50
☐	**Enclosure and fittings**	IP54 (IP65 outdoors), DIN rail, terminals, fuses or small breakers for the meters' voltage taps, glands	1	80 to 150
☐	**Cable and labels**	Shielded twisted pair for the RS-485 line, network patch cords, ferrules, labels	1 lot	40 to 80
☐	**Cellular router**	4G, DIN rail, data SIM. *The likelier case in a field.*	0 or 1	150 to 200
☐	**Soil moisture probe**	RS-485 Modbus, with a published register table, one for each zone	1 for each zone	40 to 120 each
☐	**Temperature and humidity sensor**	RS-485 Modbus, with a published register table	1	20 to 80
☐	**Pump run input** *(optional)*	A Modbus RTU digital input (dry contact) module, or one spare channel of a combined module. *If there is a pump to watch.*	0 or 1	30 to 60
☐	**Flow meter** *(optional)*	4 to 20 mA output. Needed for "the pump is running but no water is flowing". *If there is a pump to watch.*	0 or 1	150 to 500

About EUR 700 to 1,500. One soil probe for each zone. Ask three suppliers for the quote on the specification, not the brand, and keep the quotes. Buy and bench-test one kit before buying five. The reasons for each part, its settings and the mistakes to avoid are in the Hardware Reference (PDF 03).

The project manager brings

- ☐ A laptop
- ☐ A USB-to-RS-485 adapter (to talk to a meter alone)
- ☐ A loop calibrator (or a 4 to 20 mA source)
- ☐ A multimeter
- ☐ Screwdrivers (slotted and Phillips, small and medium)
- ☐ Wire strippers and a ferrule crimper
- ☐ A network cable tester
- ☐ Cable ties and labels
- ☐ A head torch
- ☐ Spare fuses and terminals
- ☐ A phone for photographs

You do not need a tool for work inside a live board: that is the electrician's. The USB-to-RS-485 adapter is how you set a probe's or a sensor's unit number before it is buried or mounted. The spares kit (one for each project manager) goes in the car too.

Survey

About one hour, with the grower and, if the plan needs one, the electrician. Bring this pack, a phone for photographs, a tape measure, a torch and a notebook. No tools are needed to survey, and nothing is opened.

Ask, in this order

The answers decide what is measured and what it costs. Write them on the survey form that follows.

1. Who decides, and who gets called at night? Names, mobile numbers, emails: these become **Who is told**. Each alert message carries an *I have seen this* link, so nobody needs a login to acknowledge; the people who should also look at the screens get a login of their own, and a login needs an email address.
2. What went wrong in the last twelve months that cost money or a crop? Write it in their words. It is the reason they buy.
3. **Which crop, which soil, how deep are the roots, and where in the field is typical** (not the wettest or the driest spot)? The irrigation limit depends on them.
4. Where are the water tank, the pump and its starter? How tall is the tank? What is the pipe's pressure? Is there a pump to watch?
5. Is there a network socket in the pump house, or cellular signal? (Look at your phone there.)
6. Is there a power socket for the box, on a circuit that stays on?
7. How far is the probe's place from the box? The data cable is planned at under 100 metres. If it is further, a probe on a radio gateway that speaks Modbus is the alternative; the gateway's maker must publish its register table. Ask the founder before promising it.
8. Who is the customer's electrician, and can they be there on install day?

Look at, and photograph

- The field: where each probe goes, and the weather sensor's place.
- The tank, the pump, the starter, and any existing level or pressure gauge and flow meter.
- The pump house: where the box and the cellular router would sit, dry, 30 cm free, no heat, within reach of a socket.
- The route for the data cable from the field to the box (note any crossing of power cable).

CHECK

Go if there is a way to the internet (cable or cellular), a socket that stays on, a safe place for each meter, an electrician for the board work, and a person who answers when called.

No-go, politely, and write why, if the only place for a sensor is unsafe to reach, the customer wants something switched or controlled, or there is no responsible person to receive alerts.

☐ Go ☐ No-go, because: _____

Survey form

KIT F · FARM OR GREENHOUSE WITH IRRIGATION

Field	Answer
Customer, site name and address, date	
Contact: name, mobile, email (first, second, third to be told)	
What went wrong last year (their words)	
Crop; soil; root depth (cm)	
Zones: how many probes; where each goes (typical, not wettest or driest)	
Distance from the probe places to the box (m); wired or radio gateway	
Tank: height (m), water source; pump (kW); starter location; pipe pressure (bar)	☐ No pump to watch
Sensors needed: level, pressure, flow, pump running (and where)	
Network: customer's cable or cellular; signal in the pump house	
Power for the box: socket location	
Electrician: name, mobile, day	
Photographs taken (list)	
Go / no-go, and why	☐ Go ☐ No-go

The drawings

Every drawing is for kit F and says so. They show how the parts connect, not terminal numbers: follow the maker's manual for terminal numbers.

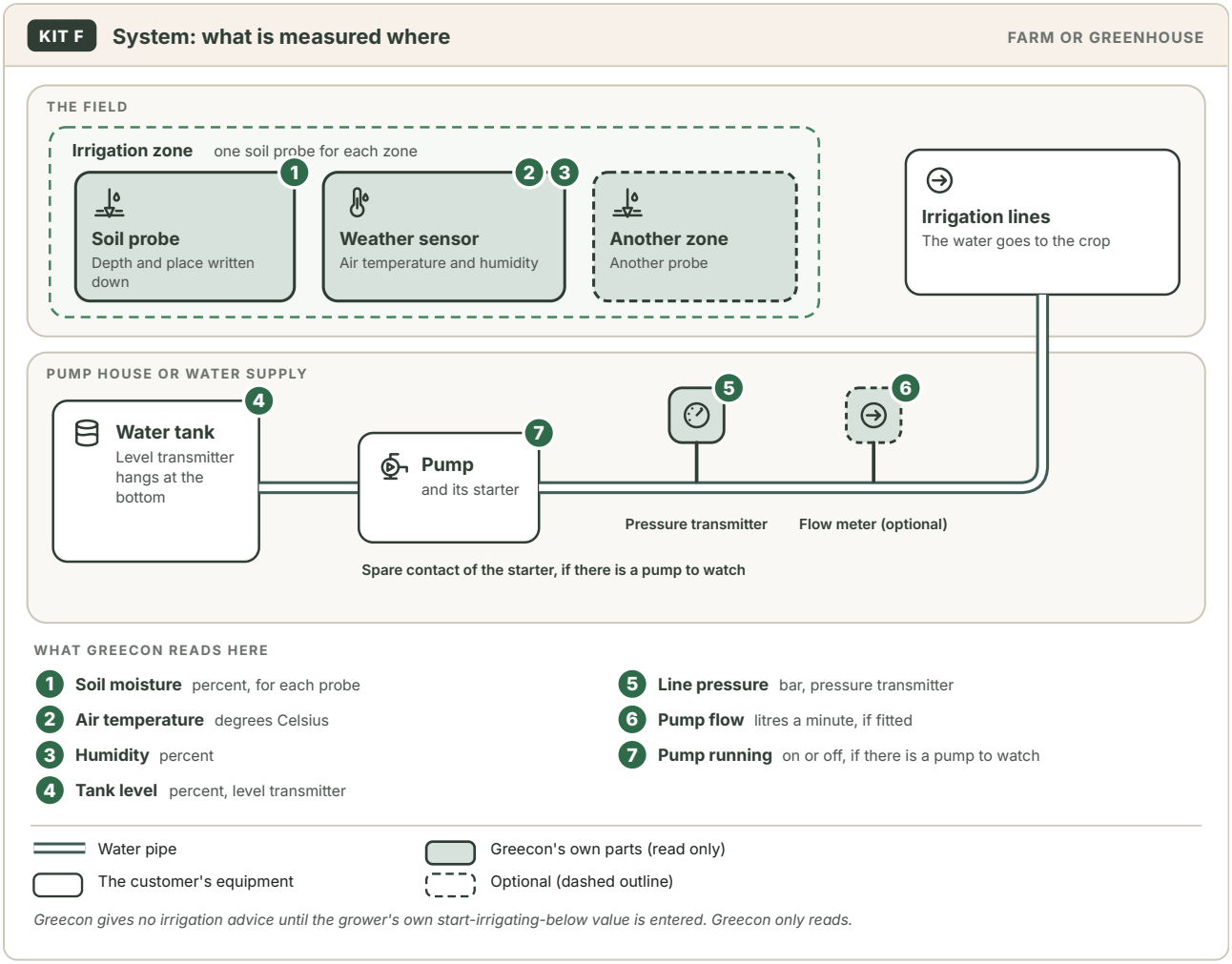


Figure F-1. Kit F, farm or greenhouse with irrigation. System: what is measured where. The dashed parts are optional.

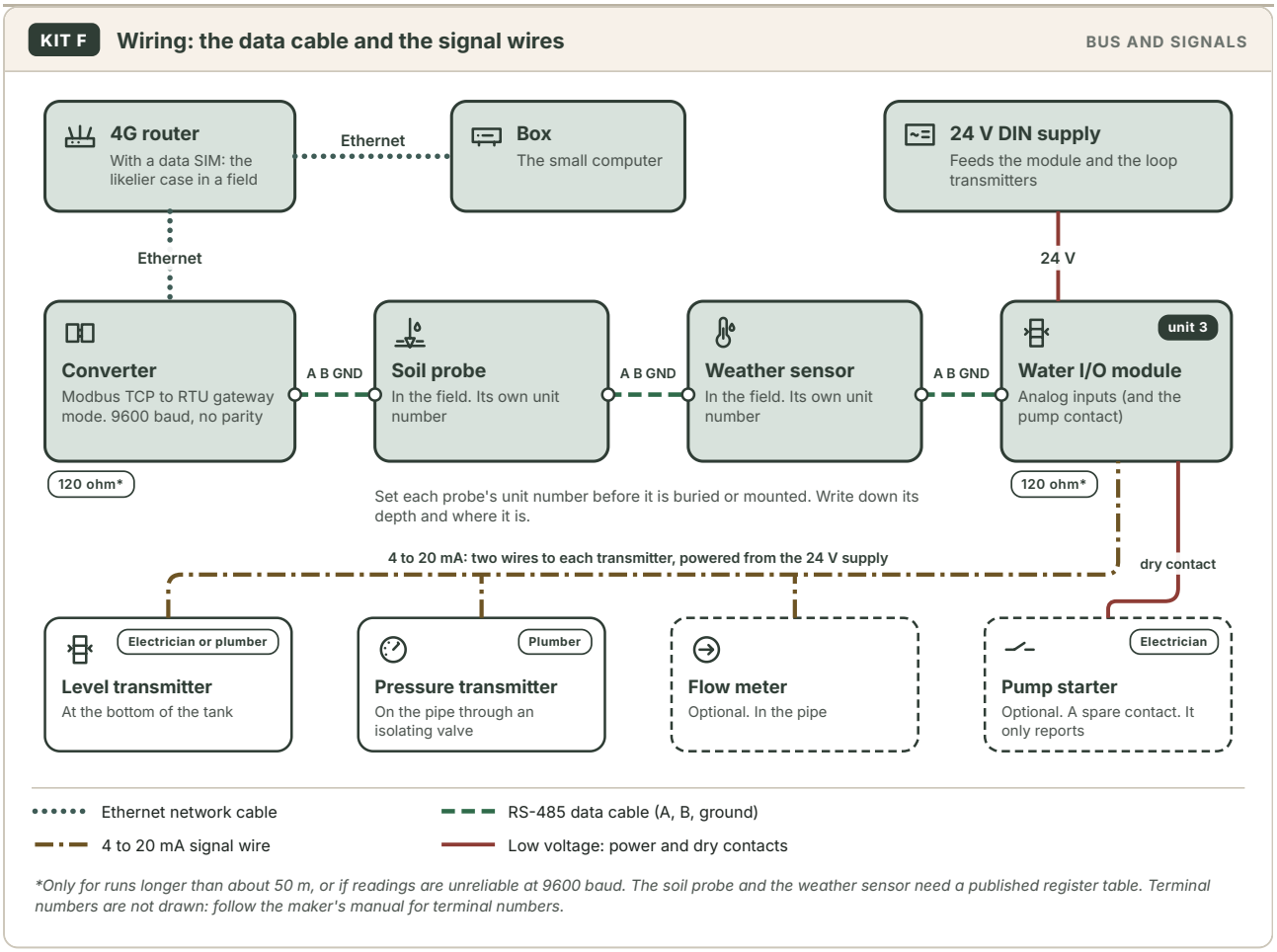


Figure F-2. Kit F. Wiring: one RS-485 cable from the converter through the soil probe and the weather sensor to the water module, with the 4 to 20 mA loops into the module.

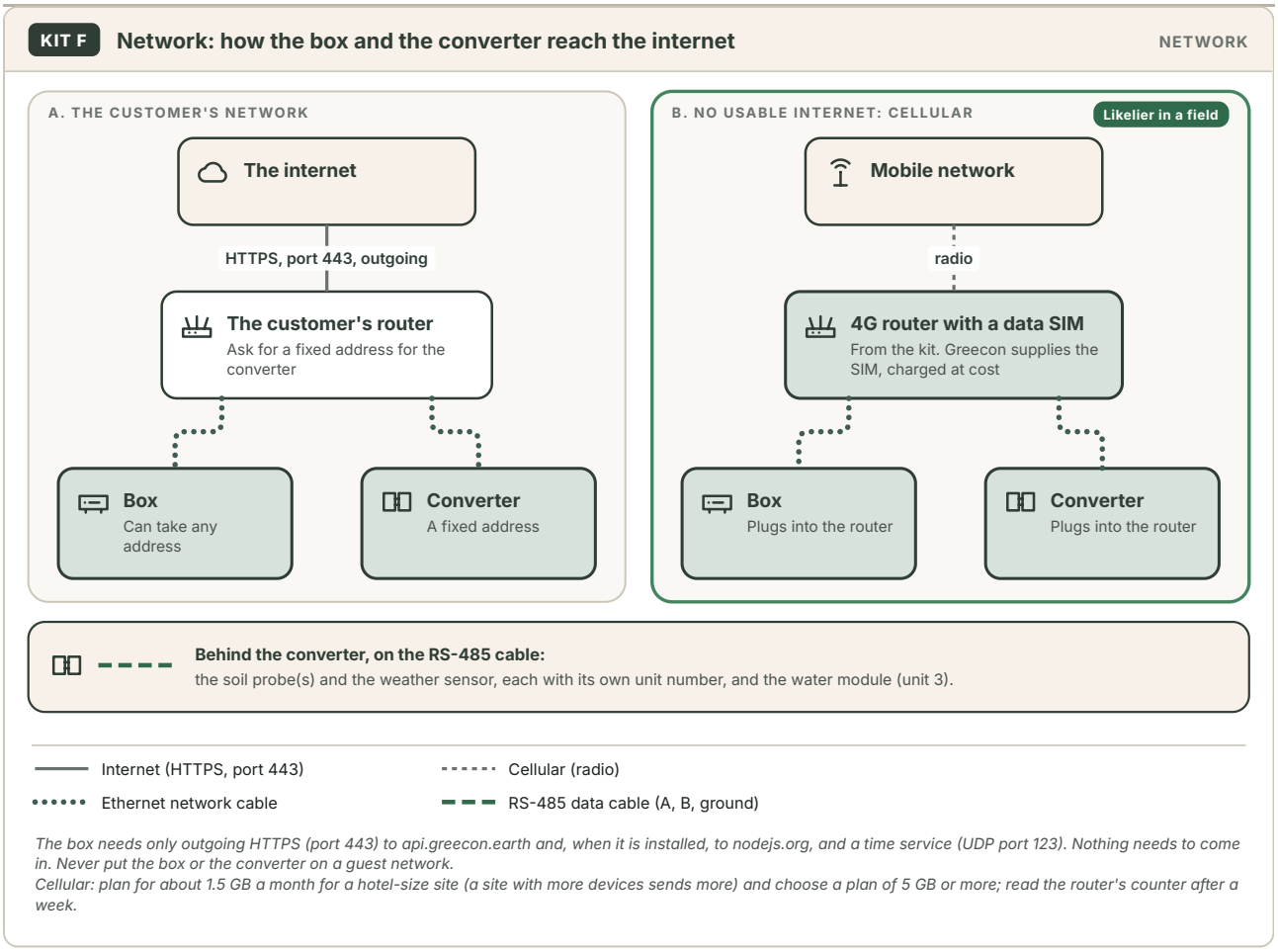


Figure F-3. Kit F. Network: a 4G router with a data SIM is the likelier case in a field; the customer's router is better where there is one.

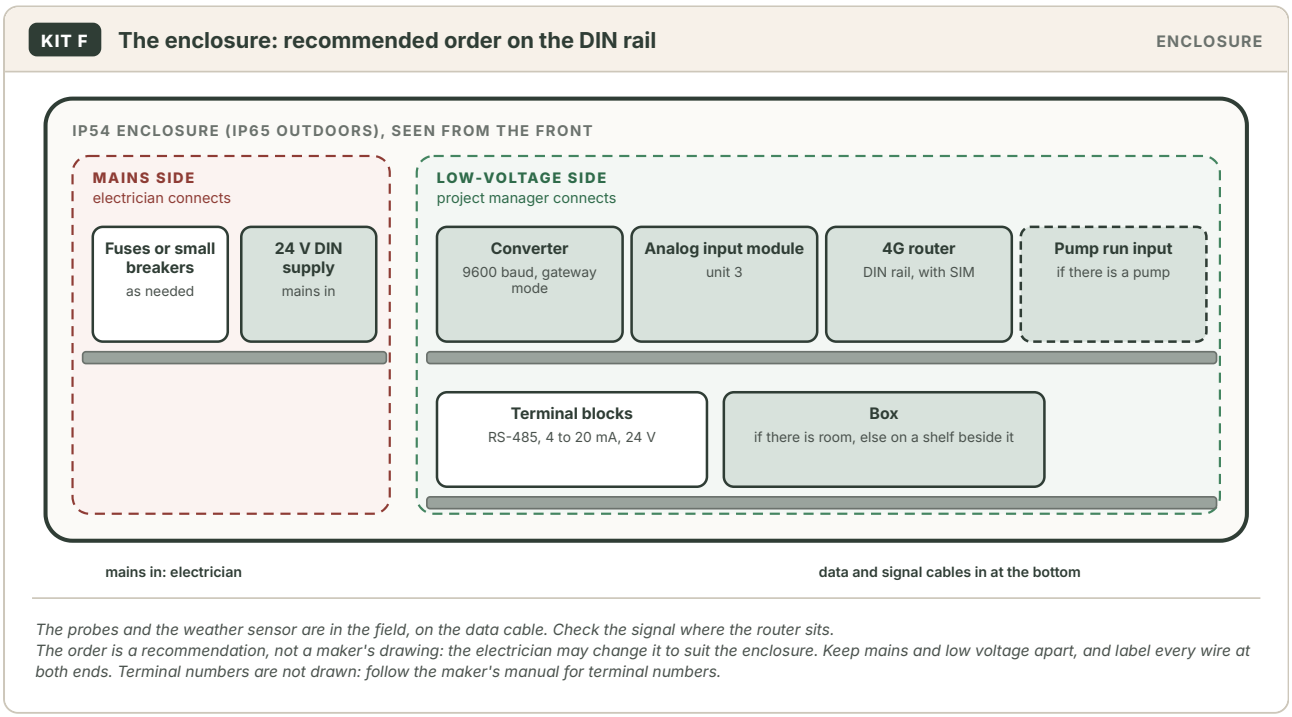


Figure F-4. Kit F. The enclosure: the recommended order of the parts on the DIN rail, mains side on the left.

The order on the rail is a recommendation, not a maker's drawing: the electrician may change it to suit the enclosure. Keep mains and low voltage apart, and label every wire at both ends.

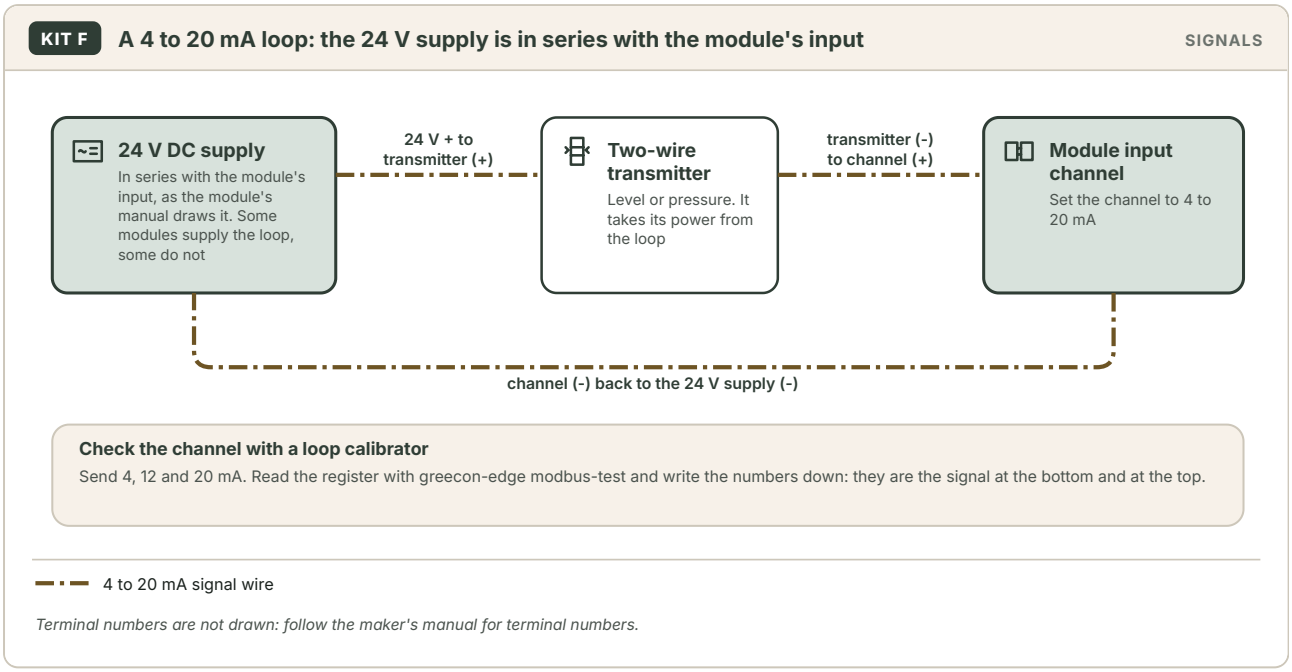


Figure F-5. Kit F. One 4 to 20 mA loop; the level, the pressure and the flow transmitter are each wired this way.

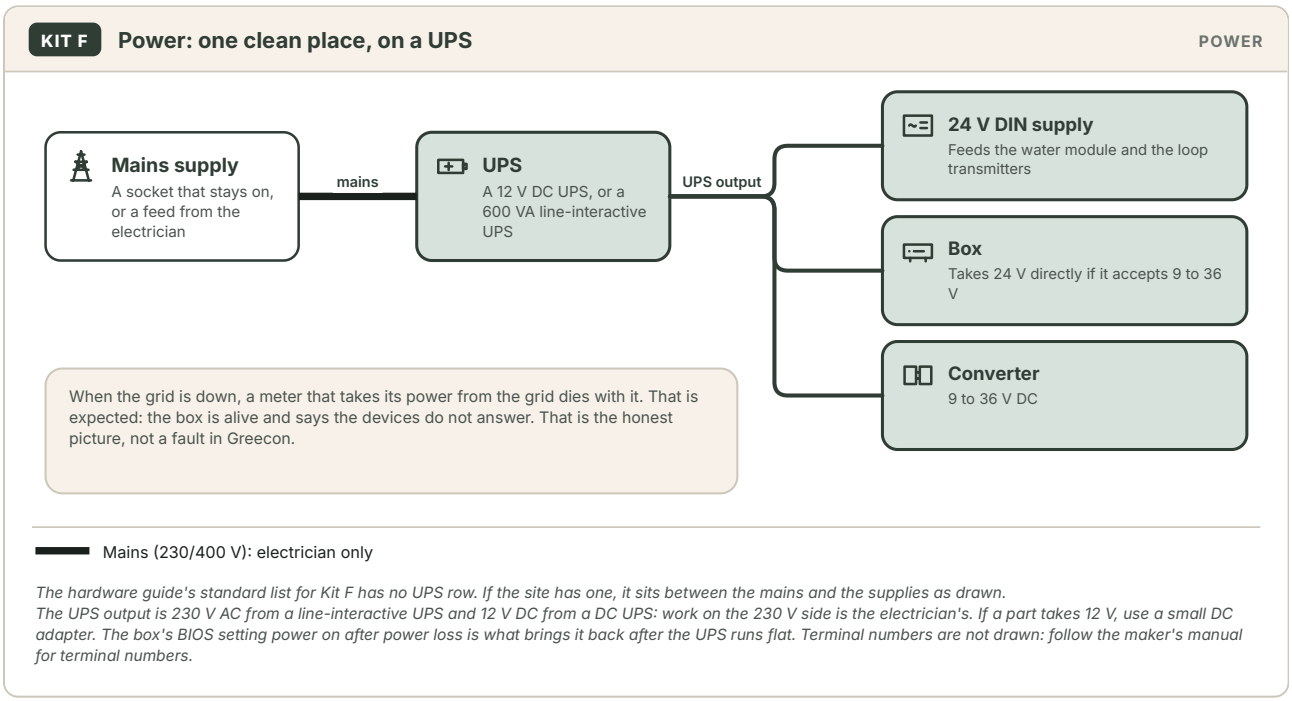


Figure F-6. Kit F. Power. The hardware guide's standard list for this kit has no UPS: decide at the survey and write it down.

At the office

Everything in this chapter is done at the office, before the site visit, and it makes install day shorter. About 30 minutes in Greecon and about two hours for the box. PROJECT MANAGER

In Greecon

- [ ] **Sign in** at `https://app.greecon.earth/login` with **Greecon team**. You land on **Today**.
- [ ] **A new customer:** if they asked through the web page, their request waits under **Requests waiting** on **Today** (and on **Requests**): press **Onboard**. Otherwise open **Clients** and press **Onboard a client** (top right): company name, owner's name, owner's email, then **Onboard client**. A box appears, **Client onboarded — save this password now**: it is the owner's temporary password, shown once. Write it on the handover page or tell it to the owner by phone. Do not send it by email. If a message says someone already has an account with that email, the person is already a customer: add the new site to the customer they already have.
- [ ] **Open the customer for setup:** in the customer's row on **Clients** press **Open for setup** (on **Today** the same button is **Open account** for a site that needs a look and **Continue setup** for a site being set up). You land on the customer's **Sites** page, with a banner saying you are there as Greecon staff, setting it up. Everything you do is recorded under your own name. The setup session lasts four hours: after that, sign in again with **Greecon team** and press **Open for setup** again. What you saved is still there.
- [ ] **Make the site:** on **Sites** press **Add a site**. In the form (*New site from a template*) choose **What kind of site**: **Farm or greenhouse with irrigation**. The page lists what the template will contain. Enter **Site name** and **Where it is**, check **Currency of the bills** (it starts as ALL) and **The site's own clock** (Europe/Tirane), and press **Create the site and start setting it up**.
- [ ] **Get the install line:** you land on the setup page, titled **Set up** and the site's name, which says `0 of N steps done`. Press **Add the box and get its install command**. A line appears, beginning `curl -fsSL https://api...` and ending in a code like `GRC-XXXX-XXXX-XXXX`. Copy it with the copy button. It is shown once. The code works once, for seven days. Press **I have copied it**.

> **STOP** **The install line is the box's key.** Anyone with the line can install a box as this site until it is used or has expired. Treat it like a password: not in email or chat, and pasted straight into the box. If it is lost, or the seven days have passed, press **Install a box, or replace this one** on the same step; it asks you to confirm and shows a new line. The old line then stops working, and the box that uses the new line takes the place of the box now installed.

The box on your desk

- [ ] **Install the operating system** on the box: Debian 12, Ubuntu 22.04 or later, or Raspberry Pi OS 64-bit, with the SSH server on, one user with a long password, and the time zone set to Europe/Tirane. The twenty-minute recipe and the BIOS settings are in the Hardware Reference (PDF 03), section 4.3.
- [ ] **Connect the box** to your office network and open a terminal on it (a keyboard and screen, or SSH).
- [ ] **Paste the install line exactly.** It runs in seven numbered steps and tells you each one: checking the box (and that it can reach Greecon and agrees with it about the time), installing Node.js, making the places the software uses, fetching and checking the Greecon software, giving the box its own credential, starting the service, and waiting to hear that Greecon can see the box. It finishes with `Done. Greecon can see this box.`

- [ ] **Refresh the setup page. Install the box** now says **Done**, with the box's name, its version and "Heard from N seconds ago". You did not press anything to make it so: the platform saw the box.
- [ ] **Label the box** with the site name and the date, and pack it. The box keeps its credential when it is moved: unplug it, carry it to the site, plug it in, and it reports from the new place without anything being typed again.
- [ ] **Pack the rest** of the kit from the shopping list, the tools, the spares kit and the printed pack.

TIP

If the line stops with `STOP:` it says what to check and how. If it stopped before step 5, *Giving this box its credential*, fix the cause and run the same line again: the code has not been used yet. From step 5 on, the code is used up, and the same line would stop with "Greecon did not give the software to this box". Then run the line again with no code at the end: `curl -fsSL https://api.greecon.earth/edge/install.sh | sudo bash -s --`. That repairs the box and keeps its credential. If it answers "This box has no code and has not been installed before", the box never got its credential: press **Install a box, or replace this one** and use the new line. The commonest stops are a box that cannot reach Greecon (no internet, a firewall that blocks outgoing HTTPS on port 443, or a clock so far out that secure connections fail), and a code already used or older than seven days. A clock only minutes out does not stop the install: it prints a warning, and the setup page later says its clock is N minutes fast.

TIP

The first time, rehearse the whole job on a desk with a laptop standing in for the meters: the Project Manager Field Manual (PDF 02), section 3.11. It takes an afternoon and finds the mistakes in the room instead of at the customer's.

Before the kit leaves the office, set the unit number of each soil probe and of the weather sensor, with the USB-to-RS-485 adapter and the maker's instructions, and write them down: **set them before the probe is buried or the sensor is mounted.** Make every one different from the others and from the water module (3).

Install day

About four to six hours. The electrician and the project manager work at the same time, in this order, so that nobody waits. The left edge of each row is black for the electrician, green for the project manager and sand for the customer.

	#	Who	Do this
☐	1	ELECTRICIAN · PROJECT MANAGER · CUSTOMER	**Walk the site together,** with the grower. Agree where each probe and the weather sensor go (typical of the zone, not the wettest or driest spot), and where the enclosure sits. Photograph everything before you start.
☐	2	CUSTOMER	Show the way to the pump house, the tank and the starter. Have a socket that stays on ready.
☐	3	ELECTRICIAN	**Apply the customer's lock-out** on the circuit that feeds the pump house.
☐	4	ELECTRICIAN	**Connect the mains to the 24 V DIN supply** in the enclosure, as its manual shows, with a fuse or small breaker as needed.
☐	5	ELECTRICIAN	**If there is a pump to watch:** connect the pump's run contact (a spare auxiliary contact of the starter) to the digital input of the water module. Never to a control circuit: the contact only reports.
☐	6	ELECTRICIAN	**Fit the level and pressure transmitters** (and the flow meter, if there is one) with the customer's plumber for the wet parts. The level transmitter hangs at the bottom of the tank and must be rated for drinking water. The pressure transmitter goes on the pipe through an isolating valve.
☐	7	PROJECT MANAGER	**Mount the enclosure** and in it the converter, the water module, the 24 V supply, the 4G router (and the box, if there is room, otherwise on a shelf beside it). Figure F-4.
☐	8	PROJECT MANAGER · CUSTOMER	**Place the soil probe(s)** at the depth and in the place agreed with the grower, in good contact with the soil, as the maker says; mount the weather sensor. **Write down each probe's depth and where it is, and photograph it.**
☐	9	PROJECT MANAGER	**Run the RS-485 data cable** from the converter to the soil probe(s) and the weather sensor, and on to the water module: one cable, a daisy chain, no branches. Shielded twisted pair, the shield joined to ground at one end only, away from power cables. Figure F-2.
☐	10	PROJECT MANAGER	**Wire the 4 to 20 mA signals:** two wires from each transmitter to the water module, with the 24 V supply in series with the module's input, as the module's manual draws it. Figure F-5.
☐	11	PROJECT MANAGER	**Set the water module:** unit number 3, and every channel you use to 4 to 20 mA.
☐	12	PROJECT MANAGER	**Set the converter:** Modbus TCP to RTU gateway mode, 9600 baud, 8 data bits, no parity, 1 stop bit, TCP port 502, and a fixed address on the site's network. Write the address down and photograph the settings page.

	#	Who	Do this
☐	13	PROJECT MANAGER	**Connect the network:** put the SIM in the 4G router, check the signal where it sits, and plug the box and the converter into its sockets (or into the customer's router if the pump house has one). Figure F-3.
☐	14	ELECTRICIAN PROJECT MANAGER	**Photograph before closing:** the wiring, the converter's settings, each device's label. Then close the enclosure.
☐	15	ELECTRICIAN	**Remove the lock-out** only when you both agree the work is closed up.
☐	16	PROJECT MANAGER	**Power up** the 24 V supply, the converter, the router and the box.
☐	17	ELECTRICIAN	**Stay until commissioning is done.**
☐	18	PROJECT MANAGER	**Open the customer for setup again** on your laptop (the four hours are over by now) and go down the platform's setup list.

Addresses and registers

Devices behind one converter share its IP address and differ only by unit number. The launch documents fix only the unit numbers 1 to 4 (grid 1, solar 2, water module 3, optional load or pump meter 4); the probes and the weather sensor need their own, chosen and written down. A suggestion, if you have none: weather sensor 5, soil probes 6, 7 and so on. Write the numbers here as you set them; you type them into the setup list in step 3.

Device in Greecon	Make and model	Unit number	IP address (the converter's)	Port
Soil probe (zone 1)				502
Soil probe (zone 2, if any)				502
Weather sensor				502
Water I/O module		3		502

Registers: all of them from the makers' manuals

There is no ready-made profile for a probe or a sensor: each reading is mapped by hand from the maker's register table (step 4 of the setup list). Under each reading's name press **Choose the register**. The register kind is **Input** for a manual's 3xxxx and **Holding** for 4xxxx; the address is counted from zero (30053 is address 52). Write what you enter:

Reading	Register kind	Address	Kind of number	How it becomes the reading
Soil moisture (zone 1)				
Air temperature				
Humidity				

For the water module's four readings (tank, pressure, flow and pump running), use the worksheet below. The module's own manual gives what the register holds at 4 and at 20 mA: check those two with a loop calibrator (send 4, 12 and 20 mA, read the register with `greecon-edge modbus-test`) and write what you read.

Reading	Register kind	Address	Conversion	Register at 4 mA	Register at 20 mA	Reading from bottom to top
Tank level			A straight line			0 to 100 (rule below)
Line pressure			A straight line			0 to the range, in bar
Pump flow			A straight line			0 to the range, in litres a minute
Pump running			Multiply and add			on or off

When the transmitter's range is longer than the tank is high: reading at the top = range ÷ tank height × 100. A 0 to 5 m transmitter in a tank 3 m high: 5 ÷ 3 × 100 = 166.7. Delete the readings the site does not have (no flow meter, no pump to watch): open **Equipment and readings**, find them under **Readings** and press **Remove**. A reading with no register holds the setup list at step 4.

The platform's setup list

The platform keeps its own list for each site. Open the site's setup page (**Set up** and the site's name). It shows up to ten steps, always in this order; a step that does not apply to the site is left out. Each step is **Done**, **To do**, **Waiting** for another step, or **Needs attention**, and the banner at the top, **Do next**, names the step to do next. The platform works most steps out by itself when it sees them work: you press buttons only for what a person must check. A step opens to show its detail and stays as you leave it.

	Step	Title on the setup list	Done by
☐	1	**Prices from the customer's bills**	Project manager (optional)
☐	2	**Install the box**	Project manager
☐	3	**Where each device is on the network**	Project manager
☐	4	**Which register holds which reading**	Project manager
☐	5	**Every device answers the box**	The platform, by itself
☐	6	**Readings arrive and look right**	The platform, by itself
☐	7	**Someone is told when the site needs attention**	Project manager
☐	9	**Alert limits agreed with the customer**	Project manager
☐	10	**Commissioned**	Project manager

Below, each step says what to do, what you should see, and what to do if you do not.

Not on this kit. The platform leaves out a step that does not apply to the site, so the list for kit F has no step 8 (Grid meter direction checked). The steps keep their order; the count on the setup page is out of the steps that do apply, not out of 10.

Step 1 Prices from the customer's bills

PROJECT MANAGER *optional*

DO	Open **Enter prices and baseline** (once something is entered it is called *Prices and baseline*; the same form is on the site's page). Enter the currency; the electricity price per kWh from the latest bill; what is paid for energy sent to the grid (leave it blank if nothing is paid); the water price per m³; a typical month before (the kWh bought and the m³ of water, from last year's bills); where those numbers came from; whether the site cannot send energy to the grid (tick it if the plant is zero-export); and the time zone, `Europe/Tirane`. Press **Save**.
YOU SHOULD SEE	The step turns **Done** and lists the prices, for example *Electricity: N ALL per kWh* (and *Water: N ALL per m³* where the site has a water flow reading).
IF YOU DO NOT	It stays **To do** and says *Not set: Greecon shows kWh but cannot say what they are worth.* That is allowed: this step is optional and does not hold up commissioning, and without a price Greecon shows kWh only. Without the typical month before, no comparison is shown, which is correct.

Step 2 Install the box

PROJECT MANAGER

DO	Press **Add the box and get its install command** and copy the line it shows (shown once, works once, lasts seven days). Run it on the box exactly as shown. This is done at the office, before install day.
YOU SHOULD SEE	**Done**, with the box's name, its version and *Heard from N seconds ago.* You do not press anything to make it so: the platform saw the box.
IF YOU DO NOT	- *Waiting for the install command to be run on the box.* The line has not been run yet. - *The box has its credential but has not reported yet.* Wait a minute and check the network cable. - *Last heard from N ago: it is offline.* Check the box's power, the network cable and the site's internet. On the box: `greecon-edge status` and `journalctl -u greecon-edge -n 50 --no-pager`. - *Online, but its clock is N minutes fast* (or slow). Run `sudo timedatectl set-ntp true`.

Step 3 Where each device is on the network

PROJECT MANAGER

DO	Under each device, enter its **IP address** (the converter's address), its **Port** (502 unless the converter's page says otherwise) and its **Unit number** (the number you gave the device, 0 to 247). Devices behind one converter share its IP address and differ only by unit number. Press **Save** under each. The box picks up a change within a minute.
YOU SHOULD SEE	**Done**: *Every device has an address,* and each device reads like *192.168.1.50:502, unit 1*.
IF YOU DO NOT	*N devices still need an address.* A wrong address is refused in words, for example *The address must be an IP address such as 192.168.1.50, or a host name.*

Step 4 Which register holds which reading

PROJECT MANAGER

DO	For equipment Greecon knows (the Eastron meters in the template) the registers are already filled in: check them against the meter's manual only if a reading looks wrong. For anything else, press **Choose the register** under the reading's name and fill in: the register kind (**Input** for a manual's 3xxxx, **Holding** for 4xxxx), the address (counted from zero, so 30053 is address 52 and 40001 is address 0), the kind of number (and, for two-register numbers, which half comes first), how it becomes the reading (**Multiply and add**, or **A straight line, such as 4-20 mA**) and which part of the value. Press **Save the register**.
YOU SHOULD SEE	**Done**: *Every reading is tied to a register.*
IF YOU DO NOT	*N readings not tied to a register yet*, each marked *No register chosen.* A reading the site does not have (a flow meter that was not fitted, a battery that does not exist) holds the list at this step: open the site, find it under Equipment, in Devices or Reading points, and press Delete.

Step 5 Every device answers the box

THE PLATFORM

DO	Nothing to press: the platform does this one by itself once the box is online and each device has its address and registers. Wait a minute.
YOU SHOULD SEE	Each device shows **Done**: *Answering (N readings)*. On the box, `greecon-edge status` lists each device as **ok**.
IF YOU DO NOT	The device shows **Needs attention** with the reason in words, for example *no answer from 192.168.1.50:502 (unit 2)*. Look in this order: a wrong unit number (the commonest), a wrong speed or parity at the converter, A and B swapped on the data cable, the converter not in gateway mode, a wrong IP address. Nothing answers at all: the IP address or port is wrong, the converter is off, or it is on another network than the box. Some readings work and others do not: a register address is wrong for that device. Test one register from the box with `greecon-edge modbus-test` (see the box commands): if it prints *Could not read*, the problem is the wiring or the settings, not Greecon.

Step 6 Readings arrive and look right

THE PLATFORM

DO	Nothing to press. Look at each reading: is it a number you believe?
YOU SHOULD SEE	Each reading shows **Done** with its value and age, for example *N %, 4 seconds ago*, and the step says *Every reading is arriving and could be real.*
IF YOU DO NOT	*No reading yet*: wait a minute. *Last reading N ago. It has stopped arriving*: a device or the converter lost power, or a cable came loose. A value that cannot be real says so, for example *237 cannot be a tank level (0 to 100). Check the register address, the scale or the sensor.* That is almost always a wrong scale, a wrong register or a wiring fault.

Step 7 Someone is told when the site needs attention

PROJECT MANAGER

DO	Open **Who is told**. **Add a person** for each name from the survey, in the order they should be told (each needs an email, a mobile number with its country code such as `+355 69 123 4567`, or both; give everyone an email address as well as a number). Press **Save the list**. Press **Send a test message** beside each person and ask them whether it arrived: in the inbox, not in spam; on the phone. Tell each person that every alert message has an *I have seen this* link: one tap, no sign-in, and it stops the message going on to the next person. Give a login of their own (the Operator role) only to those who should also look at the screens. Then, back on the setup list, press **I checked this**.
YOU SHOULD SEE	After the test: *A test message was sent N ago. Greecon cannot tell whether it was read.* After you press the button: **Done**, *A test message reached the person on the list.* The button stays grey until the list has someone on it and a test has been sent.
IF YOU DO NOT	▪ *This Greecon platform is not set up to send messages:* tell the founder; the email service must be switched on before a site can be commissioned. ▪ *Nobody is set to be told:* add the people. ▪ *Nobody on the list has an email address* (or a phone number): add one for each person. ▪ A message that did not arrive: look at **What was sent** under **Who is told**, check the address and the country code, and ask about spam. ▪ *Test messages are limited to 6 an hour.* Wait and try again.

Step 9 Alert limits agreed with the customer

PROJECT MANAGER

DO	The step lists each reading that has limits, with what applies. Go through them with the customer and agree numbers they would act on. To change one, open the **Standard** drop-down beside the reading (it says *1 set*, *2 set* once changed), type the numbers and press **Save limits**. When they are agreed, press **I checked this**.
YOU SHOULD SEE	Each reading with limits is listed with its numbers, for example *Tank level: Critical below 15 % (standard); Low below 35 % (standard).* After you confirm: **Done**, *Checked on site.*
IF YOU DO NOT	The standard limits apply until you change them: tank **Critical below 15 percent** (an Act now alert) and **Low below 35 percent** (a Check today alert); pump **No flow at or below 0.2 litres a minute** (an Act now alert once it has held for 30 seconds, raised only when the box reads both a flow and the pump's running state); battery **Low below 25 percent**. **Start irrigating below** has no standard value: Greecon gives no irrigation advice until the grower's own value is entered.

Step 10 Commissioned

PROJECT MANAGER

DO	Greyed out until every other step is done (optional steps do not hold it). Then it says: *On site, compare what Greecon shows with what you can see: the meter's own display, the tank, the pressure gauge. When they agree, confirm here.* Do the commissioning tests in this pack, write the pairs of numbers on the record, then press **Commission this site** with a note.
YOU SHOULD SEE	The progress line reads **Commissioned.**, your name and the date are on the handover sheet, and **Open the handover sheet** appears.
IF YOU DO NOT	While it is grey it says *N steps above must be done first.* and *Finish the steps above.* Finish them first.

Commissioning

Commissioning is checking, once, on site, that what Greecon shows is what the real world is. PROJECT MANAGER with the electrician and the owner's own person.

- Compare in the order of the table, and write the pair of numbers (what Greecon shows, what you can see) on the record.
- **Do not start, stop or open any equipment yourself.** The owner's own person does, because they know what else it affects.
- A check that fails is not commissioned: find the cause (the three most likely problems are at the end of this pack), fix it and check again. When every line passes, press **Commission this site** on the setup list (step 10) with a note. The progress line then reads **Commissioned.**, and your name and the date are on the handover sheet.

DO

The internet cut, step by step. Cut only the site's internet, for ten minutes: unplug the router's uplink cable (the one that goes to the wall or the modem), or take the SIM out of the cellular router. **Leave the box and the converter plugged into the router:** if the box cannot reach the converter it reads nothing, and there is nothing to test. While it is out, `greecon-edge status` shows a queue. Then restore the internet. A few minutes after it returns the queue is empty and the readings from the gap are in the downloads. **Write the result down:** it is the evidence for the statement that the box keeps its readings.

Check and how	Pass when	Greecon shows	Real	Result
Soil probe. In the air, then in a glass of water, then in the soil.	Three different readings: low in the air, high in the water, plausible in the soil			☐ Pass ☐ Fail
Weather sensor. Against a hand-held thermometer at the sensor. (This check is added by the pack: the launch documents give no tolerance, so write both numbers and judge.)	Plausible for the time of day			☐ Pass ☐ Fail
Tank level. A dipstick or the sight gauge, at two levels.	Within 5 percentage points			☐ Pass ☐ Fail
Pressure. Against the pressure gauge on the same line.	Within 0.2 bar			☐ Pass ☐ Fail
Flow (if there is a flow meter). A bucket test (fill a measured volume, time it), or the flow meter's own display.	Within 10 percent			☐ Pass ☐ Fail
Pump running (if there is a pump to watch). The owner starts the pump.	Greecon shows running within 30 seconds, and changes both ways			☐ Pass ☐ Fail
4 to 20 mA channels. With a loop calibrator, send 4, 12 and 20 mA.	The reading at the bottom, the middle and the top			☐ Pass ☐ Fail

Check and how	Pass when	Greecon shows	Real	Result
A test message. Each person on the list.	It arrived, in the inbox and on the phone			☐ Pass ☐ Fail
Power cut. Unplug the box for 30 seconds, plug it in.	Back reporting within 3 minutes, and the box says nothing about readings lost			☐ Pass ☐ Fail
Internet cut (the first install, and any time the box or its software is new). See the box above.	A queue while it is out; empty a few minutes after it returns			☐ Pass ☐ Fail

Signed off by: _____ Date: _____

Handover

About 30 minutes at the customer's, with the owner and the owner's people who are told. PROJECT MANAGER

- ☐ **Print the handover sheet.** On the setup page open **Open the handover sheet**, check it, and print it (**Print or save as PDF**; the printed page leaves out the menus). It names the customer and the site, says where each device is (address, unit number, registers), who is told and how to reach them, what was commissioned and by whom, and how to reach Greecon (`info@greecon.earth`).
- ☐ **Show the owner, in ten minutes, in this order:** the **Overview** (is everything all right, in one sentence); the site's own page under **Sites**, with the soil moisture, the air temperature and humidity, the tank level and the pressure; **Who is told** (under **Settings**) and a live test message arriving on their phone, and that each message carries an *I have seen this* link, so that nobody needs a login to acknowledge an alert; **Reports**, under **Downloads**.
- ☐ **Write the probe's depth and place** on the handover record, and say that soil readings are only as good as the contact between the probe and the soil.
- ☐ **Say that Greecon gives no irrigation advice yet,** and agree when you will ask what reading means "time to water" (after the first week).
- ☐ **Ask at what air temperature frost damages the crop, and in which weeks.** Enter it as the frost limit; say that the alert is an Act now message, possibly in the night, and agree who is woken.
- ☐ **Say the two things in the boxes below, in these words:** what Greecon is, and how an alert reaches people.
- ☐ **Leave** the printed sheet, signed; the box's label; and a card with `info@greecon.earth` and your phone. **File** the photographs and the survey form in the shared drive, in a folder named for the customer.
- ☐ **Leave the customer's account:** press **Back to the Greecon console** (the banner's button). It returns you to **Today**.

SAY

Greecon watches your equipment and tells you. The protection built into your equipment remains the safety layer. You will get one message each morning. If there is nothing to do, it says so.

SAY

How alerts reach people: **Act now** *goes by email at once and by text ten minutes later if nobody has acknowledged it; thirty minutes after the alert began, the next person is told by email and text, and so on down the list. Saying you have seen it means pressing the I have seen this link in the message (no sign-in, good for 72 hours) or* **Acknowledge** *on the* **Alerts** *page (for a person with a login).* **Check today** *is one email, held until morning if it is night. Beyond those, three short notes only: one to the people who were told when somebody acknowledges the alert (who took it), one when the problem has stayed clear for 15 minutes (***Resolved***), and one morning message at 07:30 saying whether everything is all right. Quiet hours (22:00 to 07:00) hold everything except Act now.*

STOP

Never promise a saving, a percentage or "AI". Say what Greecon measures and shows. The product shows what the customer's own solar kept on site, at the customer's own prices, from real readings. That is the claim. And say that Greecon is not a safety system.

Their data is theirs. Every reading for the last 31 days, hourly averages for up to a year, the alerts, the audit log and how the account is set up can be downloaded whenever they want, from **Reports** under **Downloads**. If they stop, they can ask Greecon for every reading it holds, as a file.

Handover record

KIT F · FARM OR GREENHOUSE WITH IRRIGATION

Field	Answer
Customer and site	
Date, and who was there	
Who is told, in order (name, mobile with country code, email)	
Each probe: depth, and where it is	
Alert limits agreed (tank Critical below, Low below; no-flow level if a pump is watched)	
“Start irrigating below”: not set yet; to be asked after the first week, on	
Setup list: every step Done and the site Commissioned	☐ Yes
Handover sheet printed and signed	☐ Yes
Shown: Overview, the site’s page, Who is told with a live test message, Reports	☐ Yes
Left: label on the box; card with `info@greecon.earth` and your phone	☐ Yes
Photographs and survey form filed in the customer’s folder	☐ Yes

Project manager: name and signature

Owner: name and signature

The three most likely problems

What you see	Likely cause	What to do
A probe or the weather sensor does not answer	A wrong unit number (many are delivered as unit 1, the same as another device), a speed or parity that does not match the converter, A and B swapped, no power	Check each in that order. Test the register from the box (`greecon-edge modbus-test`) against the maker's table. A device whose number was changed before burying is easy to check; one that was not must be dug up, so set every number before the probe goes in the ground
The soil reading looks wrong (constant, or the same in air and in water)	A wrong register, the wrong register kind (Input or Holding), a wrong scale, or poor contact between the probe and the soil	Re-test the register against the maker's table; repeat the air, water and soil check; check the probe's depth and contact
A frosty morning reads as a hot afternoon	The air temperature was mapped as an unsigned number	Map it as int16 (signed), with the scale of the sensor's table
No irrigation advice is shown	"Start irrigating below" has no standard value	After the first week of readings, ask the grower what reading means "time to water" and enter it as the limit (setup step 9)

More, by symptom, are in the Appendices of the Complete Deployment Guide (Appendix C) and in the Field Manual. A device that does not answer is, in this order: a wrong unit number (the commonest), a wrong speed or parity at the converter, A and B swapped on the data cable, the converter not in gateway mode, a wrong IP address.